

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
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Section / Batch:	ITA0512	Date:	19/08/2026

Code of Conduct

I **YAVANIKA S [192521258]** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Yavanika S

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Question 6 — Retail Motion Tracking Logic

A retail system requires:

- Accurate tracking when crowd density is low
- Stable tracking when crowd density is high

Observations:

Optical Flow: accurate in low density, noisy in high density

Motion Models: stable in high density, less detailed

Task: Compare both and evaluate a suitable approach or combination. Justify your reasoning logically.

1. Understanding of the Industry Problem

Retail surveillance and analytics systems must track customer movement continuously across the day.

A single, fixed tracking method cannot satisfy both extremes simultaneously, which is the core engineering constraint in this problem.

The requirement can be broken into two opposing sub-goals that must be satisfied by one unified system:

- Accuracy under low density — when few shoppers are present, the system should track each individual precisely (position, path, dwell time near shelves).
- Stability under high density — when many shoppers overlap and occlude one another, the system should not lose or swap identities, even if fine-grained accuracy is reduced.

This is a classic accuracy-versus-robustness trade-off in computer vision, and it directly motivates comparing two candidate techniques — Optical Flow and Motion Models — before deciding whether one, or a combination of both, is the correct engineering solution.

2. Application of Engineering Concepts

Optical Flow

Optical Flow estimates the apparent motion of brightness patterns between two consecutive frames by computing a per-pixel displacement vector. Algorithms such as Lucas-Kanade (sparse, feature-based) and Farneback / Gunnar-Farneback (dense, whole-frame) rely on the brightness-constancy assumption:

$$I(x, y, t) = I(x + dx, y + dy, t + dt)$$

Because it directly measures motion from pixel intensity changes, Optical Flow produces very fine-grained, accurate trajectories when subjects are clearly separated.

Comparative Summary

Parameter	Optical Flow	Motion Models
Core Principle	Per-pixel brightness displacement between frames	State prediction + correction (e.g., Kalman filter)
Accuracy (Low Density)	Very high — precise individual paths	Moderate — smoother, less granular
Stability (High Density)	Low — noisy, prone to ID switches under occlusion	High — predicts through occlusion and clutter
Computational Load	Higher (dense field computed every frame)	Lower (compact state vector per object)
Failure Mode	Breaks down when motion boundaries overlap	Loses fine detail; drifts if occlusion is prolonged
Ideal Retail Zone	Aisles, entrances, fitting rooms (sparse)	Checkout queues, sale events, escalators (dense)

Optical Flow vs. Motion Models — Head-to-Head		
CRITERIA	OPTICAL FLOW	MOTION MODELS
Low-Density Accuracy	Excellent — captures fine pixel-level motion detail	Moderate — coarser, prediction-based estimate
High-Density Robustness	Poor — noisy & error-prone under occlusion/clutter	Strong — predicts trajectory through occlusion
Computation Cost	Higher — dense per-pixel flow field every frame	Lower — compact state-space prediction (position, velocity)
Best Fit Zone	Sparse areas: aisles, entry / exit points	Dense areas: checkout queues, sale-event crowds

Figure showing side-by-side comparison of Optical Flow and Motion Models across four key criteria.

3. Solution Design — Proposed Hybrid Approach

Neither technique alone satisfies both requirements stated in the problem. Optical Flow alone would fail during crowded events (unstable), and Motion Models alone would sacrifice needless accuracy during quiet hours (over-engineered for a simple case). The engineering solution is therefore not to pick one algorithm, but to design an adaptive hybrid pipeline that switches — or blends — between the two based on real-time crowd density.

3.1 Design Rationale

- A lightweight crowd-density estimator (people count per frame or per zone) continuously monitors occupancy.
- Below a defined density threshold, the system routes frames to the Optical Flow tracker to exploit its higher precision.
- Above the threshold, the system switches to the Motion-Model tracker to preserve identity continuity under occlusion.
- A confidence-weighted fusion stage merges outputs at the boundary zone so tracking does not ‘jump’ abruptly when density crosses the threshold.

- This design mirrors how real commercial retail-analytics platforms (e.g., people-counting and heatmap systems) operate — they rarely use a single fixed algorithm, but rather a perception pipeline that adapts to scene conditions.

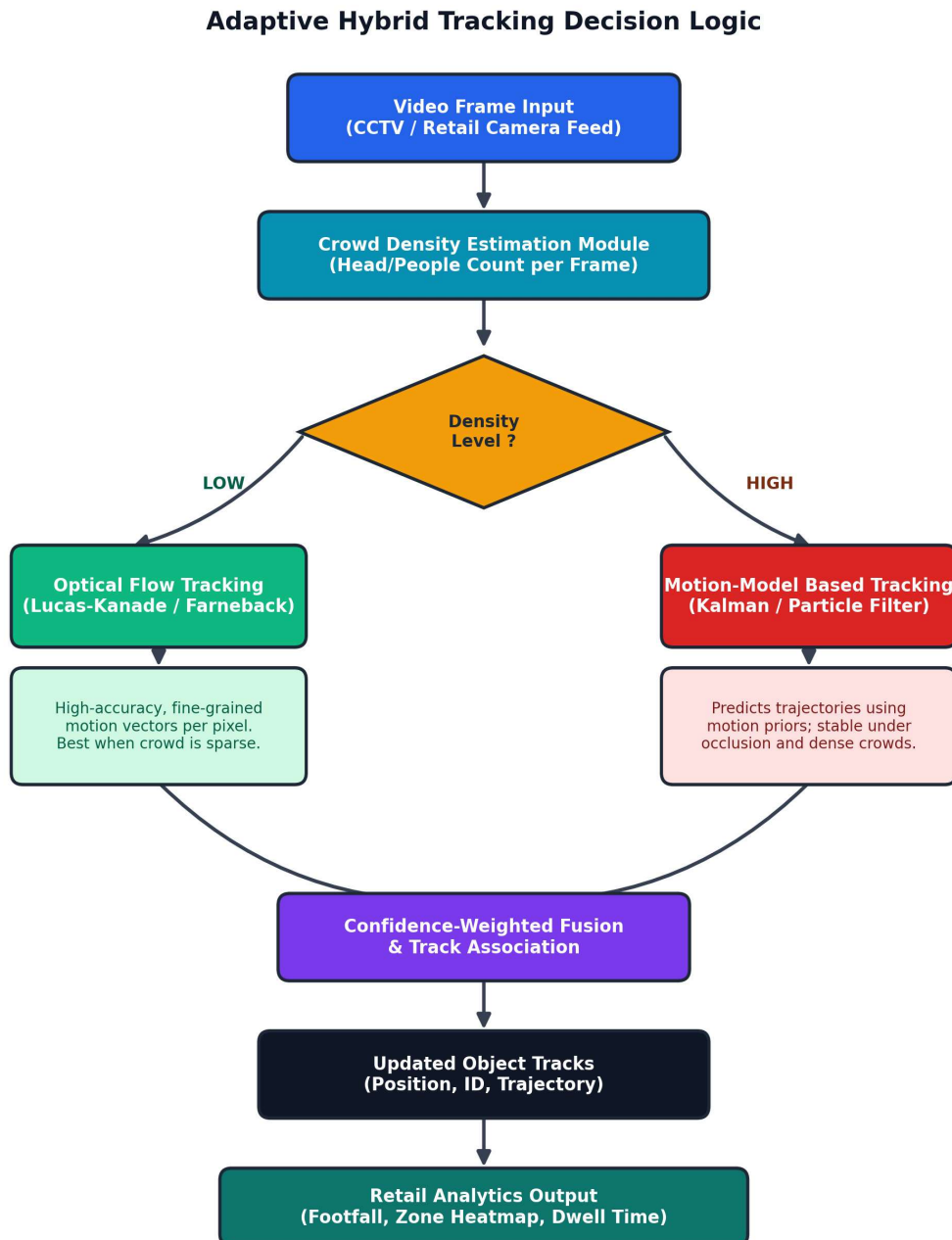


Figure 1: Adaptive decision logic showing how the system routes each frame to Optical Flow or Motion-Model tracking based on measured crowd density, then fuses the results.

3.2 Store-Wide Deployment

Applying this logic across a real store layout shows why the hybrid design is practical: naturally sparse zones (entrance, aisles, fitting rooms) default to Optical Flow, while naturally dense or occlusion-prone zones (checkout counters, promotional/sale areas, escalators) default to Motion-Model tracking. A central fusion engine reconciles both streams into one consistent set of object tracks for store-wide analytics.

Store-Wide Deployment Map of the Hybrid Tracking System

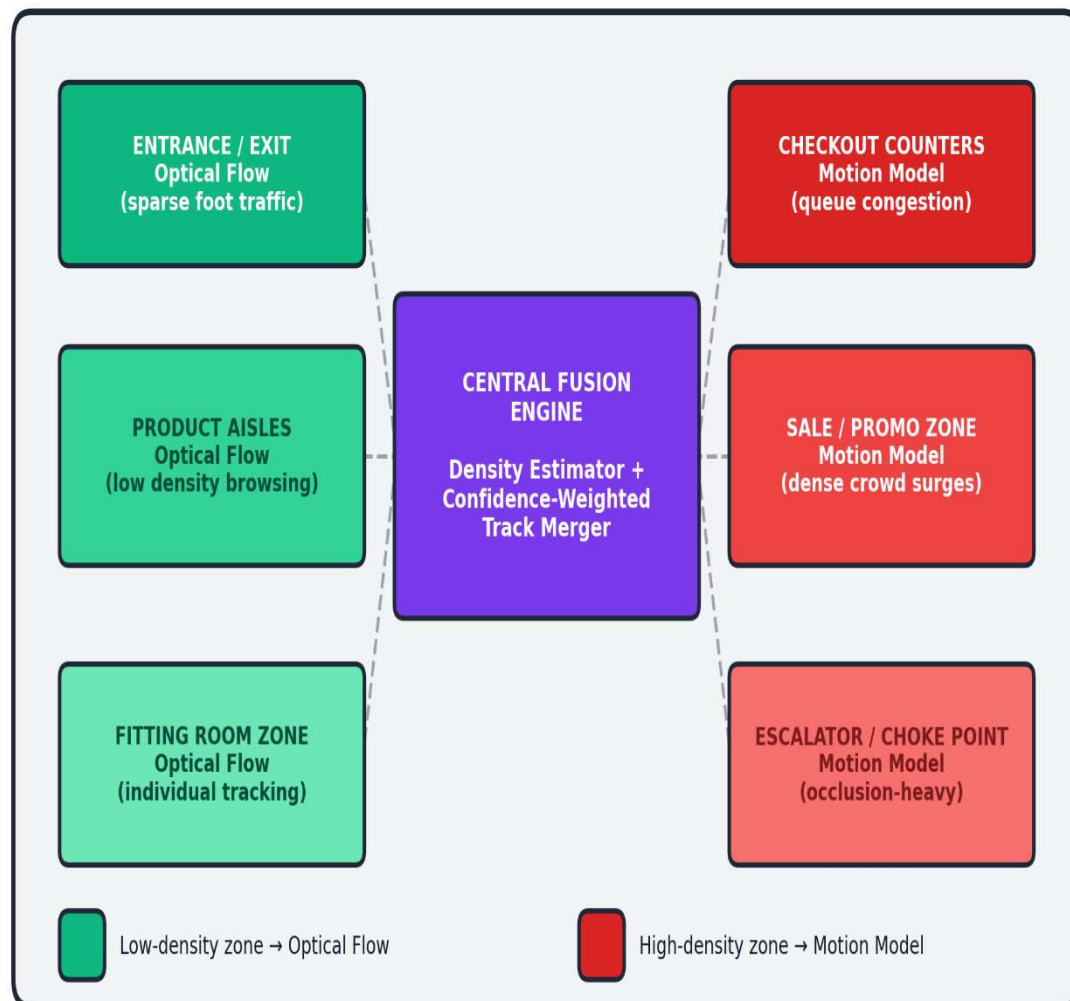


Figure 2: Zone-wise deployment of the two techniques across a typical retail floor plan, coordinated by a central fusion engine.

4. Use of Tools and Technologies

- OpenCV (Python) — `cv2.calcOpticalFlowPyrLK()` for sparse Lucas-Kanade flow; `cv2.calcOpticalFlowFarneback()` for dense flow field computation.
- `cv2.KalmanFilter` / `filterpy` library — for motion-model based state prediction and correction (position, velocity).
- Background subtraction / HOG or a lightweight CNN detector — to estimate people count for the density-estimation module.
- Hungarian algorithm (`linear_sum_assignment`) — for data association between predicted and detected positions during fusion.
- Deep-SORT style ID management — to keep consistent object IDs when handing off between the two trackers.

5. Analysis and Logical Justification

The comparison in Section 2 shows a clear complementary relationship: Optical Flow's weakness (instability under occlusion) is exactly Motion Models' strength, and Motion Models' weakness (coarser detail) is exactly Optical Flow's strength. This complementary pairing is the logical basis for combining rather than choosing one exclusively.

Quantitatively, the decision threshold can be tuned using a simple density metric — e.g., number of detected heads per square metre of camera view. A practical starting rule is:

- $\text{Density} < \text{threshold}$ (sparse) → use Optical Flow → prioritises path accuracy for behaviour analytics (dwell time, shelf attention).
- $\text{Density} \geq \text{threshold}$ (crowded) → use Motion Model → prioritises identity stability for safety and queue-management analytics.

6. Conclusion

Optical Flow and Motion Models each solve one half of the retail tracking problem but fail at the other half. By estimating crowd density in real time and adaptively routing between the two — reinforced by a confidence-weighted fusion stage — the proposed hybrid pipeline delivers accurate tracking in low-density conditions and stable tracking in high-density conditions, fully satisfying the requirements of the industry problem.